

MD MEHEDI HASAN

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OBJECTIVE

My master's research developed and validated an RFID-based Digital Twin simulation system for emergency department resource management in Bangladesh hospitals, demonstrating that dynamic backup resource reallocation maintains 99.7% alert response capability under mass casualty surge conditions. At the PhD level, I aim to extend this toward adaptive, real-time intelligent systems that integrate machine learning with RFID localization and Digital Twin frameworks for resource-constrained healthcare environments — with a focus on deployment-feasibility in developing-country settings. I am seeking a fully funded PhD position where I can contribute to both system design and real-world translational research in healthcare informatics or biomedical engineering.

EDUCATION

Saga University

Oct 2024 – Sep 2026 Saga, Japan

Master of Engineering — Computer Science & Information Technology

Supervisor: Professor Eisuke Hanada · GPA: 3.9 / 4.0 (Final semester: 4.0 / 4.0)

English proficiency: TOEFL scheduled August 2026 (target 85+)

Thesis (submitted; defense July 14, 2026): A Digital Twin Approach to Emergency Department Resource Management

This thesis proposes a two-stage Digital Twin framework for emergency department resource management, combining a MATLAB agent-based simulation system with a Unity 3D virtual reality visualization layer for RFID-based localization and resource tracking across a 20×15 m ward. Evaluated across 50 stochastic simulation runs spanning four census strata, the system achieves 99.7% alert resolution and an 83-second mean response time, with the KPI-driven Adaptive Authority engine converging nurse staffing within 4–5 simulated days. The framework establishes a reproducible, low-cost methodology for RFID-based hospital digital twins applicable to resource-constrained healthcare settings in Bangladesh and similar environments.

Bangladesh Army International University of Science & Technology (BAIUST)

2018 – 2022 Cumilla, Bangladesh

Bachelor of Science — Computer Science & Engineering

GPA: 3.16 / 4.0

Supervisors: Mohammad Asaduzzaman Khan (Assoc. Prof. & Head, CSE); Sabina Zaman (Lecturer, CSE)

RESEARCH EXPERIENCE

Hanada Laboratory, Saga University

Oct 2024 – Present Saga, Japan

Graduate Researcher

- Designed and implemented an RFID-based agent-simulation system in MATLAB (V5.3); validated across 50 stochastic runs (39–762 patient arrivals) spanning four census strata with 99.7% alert resolution and 83.0s mean response time.
- Built a Digital Twin visualization layer in Unity 3D (11 C# scripts) integrating RFID coordinate data with NavMesh-based pathfinding; achieved functional equivalence with the MATLAB 2D baseline (100% alert response coverage, 2.7x surge modulation accuracy).
- Developed and validated a dynamic backup resource reallocation mechanism — surge runs (avg. 383 patients) showed no measurable response-time degradation versus no-surge runs (avg. 131 patients), a 192% volume increase absorbed with only a 0.2-second difference.
- Conducted cross-census robustness analysis across 50 stochastic runs, identifying and resolving a stochastic dispatch anomaly (Run R13) through 12 targeted low-census simulation runs, distinguishing rare positional tail risk from systematic design failure.
- Supported by a Research Assistantship (RA) from the laboratory research grant throughout the MEng programme (¥19,800/month).

International Research Presentation — Saga University × Kasetsart University

Feb 23–27, 2026 Bangkok, Thailand

Cross-Cultural Academic Collaboration Program

- Presented RFID-based emergency department simulation research to an international academic audience as part of a bilateral collaboration program between Saga University (Japan) and Kasetsart University (Thailand).
- Participated in cross-cultural academic exchange covering research methodologies, healthcare informatics applications, and bilateral research collaboration opportunities across Southeast Asian healthcare contexts.

PATENTS AND PUBLICATIONS

C=Conference · J=Journal · S=In Submission

[C.1] Improving Patient Care in the Emergency Wards of Bangladesh Through an RFID-Based Web Simulation

Hasan MD Mebedi, Hossain Shakil, Eisuke Hanada

20th EAI International Conference on Body Area Networks (EAI BodyNets 2025), Dec 2025 (2025) [\[Accepted / In Press\]](#)

[C.2] Digital Twin-Based Emergency Resource Localization System for Crowded Hospital Wards Using RFID and VR Visualization

Hasan MD Mebedi, Hossain Shakil, Eisuke Hanada

IEICE General Conference 2026 — Session B-19, Kyushu Sangyo University, Mar 2026 (2026) [\[Published\]](#)

[C.3] Real-Time RFID Tracking Simulation for Hospital Resource Management: A 3D Unity Implementation with Dynamic Surge Response and VR Readiness

Hasan MD Mebedi, Eisuke Hanada

ISMICT 2026 — 20th Int'l Symposium on Medical Information and Communication Technology (2026) [\[Under Review\]](#)

[C.4] Cross-Census Stability and Surge Resilience of RFID-IoT Dispatch in a Bangladeshi Emergency Department Simulation

Hasan MD Mebedi, Eisuke Hanada

EAI HealthyAIoT 2026 — 11th EAI Int'l Conference on AIoT Healthcare Systems, Guangzhou, China (2026) [\[Under Review\]](#)

[S.1] An RFID-Based Dynamic Resource Management Simulation for Smart Hospital Wards: A Bangladesh Healthcare Perspective

Hasan MD Mebedi, Eisuke Hanada

Journal (Extended from EAI BodyNets 2025 — target submission July 2026) (2026) [\[In Preparation\]](#)

Co-authored contributions:

[C.5] Transfer Learning-Based Renal Ultrasound Classification for CAKUT Screening: A Preliminary Study Using ResNet-50

Ahmed Alif, Hasan MD Mebedi, Eisuke Hanada

ISMICT 2026 — 20th Int'l Symposium on Medical Information and Communication Technology (2026) [\[Under Review\]](#)

[C.6] A Lightweight AIoT Framework for CAKUT Screening in Resource-Limited Community Clinics: System Design and Edge Deployment Feasibility

Ahmed Alif, Hasan MD Mebedi, Eisuke Hanada

EAI HealthyAIoT 2026 — 11th EAI Int'l Conference on AIoT Healthcare Systems, Guangzhou, China (2026) [\[Under Review\]](#)

[C.7] Traffic-Aware Routing and Real-Time Clinical Synchronisation for Mobile Emergency Medical Services in Resource-Constrained Environments

Das Bishal, Hasan MD Mebedi, Eisuke Hanada

ISMICT 2026 — 20th Int'l Symposium on Medical Information and Communication Technology (2026) [\[Under Review\]](#)

TEACHING, MENTORING & OUTREACH

Graduate Peer Mentor, Saga University

Apr 2026 – Sep 2026 Saga, Japan

Academic & Settlement Support for International Student — Faculty-Remunerated

- Provided comprehensive peer mentoring to a newly arrived international student at Saga University, covering academic support (subject guidance in computer science and engineering) and practical settlement assistance — including official paperwork, residence registration, banking, health insurance, and navigation of daily life in Japan; remunerated by the university on an hourly basis (avg. ¥18,000/month).

Team Leader & English Presentation Coach

May – Jul 2025 & 2026 Saga, Japan

Saga University × Chienkan High School Science English Outreach Programme

- Led two consecutive teams of Japanese final-year high school students in a collaborative outreach programme between Saga University and Chienkan High School, guiding each team from initial project concept through to final scientific presentation; supported cross-cultural exchange between international researchers and students preparing for university entry.
- 2025: Guided scientific poster preparation — revised content structure, corrected English, and coached presentation delivery. 2026: Led oral presentation development — restructured project framing, updated slides and manuscript, and coached spoken English delivery for final presentation.

PROJECTS

Hospital Management System — Mainamati Cantonment General Hospital (MCGH)

Jan – May 2022, Bangladesh

BSc Final Year Project · BAIUST (Course: CSE-400)

Tools: HTML, CSS, JavaScript, MySQL

- Developed a web-based hospital management system for a real military hospital in Cumilla, Bangladesh, as part of a 5-member team; contributed as a frontend developer.
- System covered patient registration, appointment scheduling, ward allocation, and medical records — replacing manual paper-based workflows.
- Supervised by Mohammad Asaduzzaman Khan (Associate Professor & Head, CSE) and Sabina Zaman (Lecturer, CSE), BAIUST.
- Team: Thasin Mahmud, Md. Imran Hossain Emu, Md. Mabin Hossain Munna, Ahamed Jamil Bhuiyan, Md. Mebedi Hasan.

SKILLS

Proficient

Simulation & Modeling	MATLAB (Agent-Based, Monte Carlo, V5.3), Unity 3D, Digital Twin architecture
RFID & IoT	Passive UHF RFID, Active RFID, Chipless RFID, RSSI trilateration, LANDMARC
3D Visualization & VR	Unity 3D, NavMesh pathfinding, RFID pulse animation, real-time data integration
Programming	MATLAB scripting, C# (Unity), Python
Research & Writing	Academic writing, simulation design, literature review, conference presentation, LaTeX

Familiar

Web & Frontend	HTML, CSS, JavaScript
Tools & Platforms	Git / GitHub, Arduino, MS Office
Design Software	Adobe Photoshop, Adobe Illustrator, Adobe After Effects
Data	CSV pipeline, edge-cloud architecture, statistical analysis

Languages

Language	Listening	Reading	Speaking	Writing	Notes
Bengali	C2	C2	C2	C2	Native
English	B2	C1	B2	B2	Professional — TOEFL scheduled Aug 2026
Japanese	A2	A2	A2	A1	Daily conversation (approx. JLPT N4; exam not yet taken)
Hindi/Urdu	B1	A1	B1	A1	Conversational; limited reading/writing

Levels: A1/A2 — Basic user | B1/B2 — Independent user | C1/C2 — Proficient user (CEFR scale) | JLPT: N5 (beginner) → N1 (advanced)

HONORS, SCHOLARSHIPS & AWARDS

MEXT Honors Scholarship — Japanese Government Ministry of Education, Culture, Sports, Science and Technology (MEXT), Japan ¥48,000/ month · Competitive national scholarship	Oct 2024 – Mar 2025
Saga City Scholarship Saga City Government, Japan · ¥20,000/month · Competitive municipal scholarship	May 2025 – Mar 2026
Full Tuition Exemption — Saga University National university tuition fully waived (~¥535,800/year)	Oct 2024 – Sep 2026
Candidate for Academic Incentive Award IEICE General Conference 2026, Session B-19-01 — Healthcare and Medical Information Communication Technology, Kyushu Sangyo University <i>Paper: Digital Twin-Based Emergency Resource Localization System for Crowded Hospital Wards Using RFID and VR Visualization</i>	Mar 2026
Perfect Semester GPA — 4.0 / 4.0 Final completed semester (2025 Fall), Saga University	2025

REFERENCES

1. Professor Eisuke Hanada

Faculty of Science and Engineering, Saga University, Japan

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Relationship: Master's Thesis Supervisor